Testing SSD Speed

Open the terminal on your Jetson AGX Thor system.

1. View the SSD's mount information.

```
df -h
```

```
jetson@jetson-desktop:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none             16G     0   16G   0% /dev
tmpfs            16G   40K   16G   1% /dev/shm
tmpfs            16G   22M   16G   1% /run
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
tmpfs            16G     0   16G   0% /sys/fs/cgroup
/dev/nvme0n1p1  117G  5.1G  106G   5% /
tmpfs           3.2G  124K  3.2G   1% /run/user/1000
/dev/mmcblk0p1   28G   11G   16G  39% /media/jetson/1dad3c26-f494-4334-8dff-3424
ea22d47a
jetson@jetson-desktop:~$
```

/dev/nvme0n1p1 is the SSD you want to test.

2. Install hdparm.

```
sudo apt-get install hdparm
```

```
jetson@jetson-desktop:~$ sudo apt-get install hdparm
[sudo] password for jetson:
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  apt-clone archdetect-deb bogl-bterm busybox-static cryptsetup-bin
  dpkg-repack gir1.2-timezonemap-1.0 gir1.2-xkl-1.0 grub-common
  kde-window-manager kinit kio kpackagetool5 kwayland-data kwin-common
```

3. Test file read speed.

```
sudo hdparm -tT /dev/nvme0n1p1
```

```
jetson@jetson-desktop:~$ sudo hdparm -tT /dev/nvme0n1p1

/dev/nvme0n1p1:
 Timing cached reads:   10810 MB in  1.99 seconds = 5420.97 MB/sec
 Timing buffered disk reads: 4234 MB in  3.00 seconds = 1410.84 MB/sec
jetson@jetson-desktop:~$
```

4. Test file write speed. The test time depends on the final count; here, 5000 represents approximately 5GB of data. You can increase or decrease this count as needed.

```
sudo dd if=/dev/zero of=/home/jetson/test_write.img bs=1M count=5000
```

```
jetson@jetson-desktop:~$ sudo dd if=/dev/zero of=/home/jetson/test_write.img bs=
1M count=5000
5000+0 records in
5000+0 records out
5242880000 bytes (5.2 GB, 4.9 GiB) copied, 10.4812 s, 500 MB/s
jetson@jetson-desktop:~$
```

5. Delete the file

The write speed test above will generate a 5GB file named test_write.img in the jetson user directory. You need to manually delete it using the following command:

```
rm -rf /home/jetson/test_write.img
```

```
jetson@jetson-desktop:~$ ls
Desktop    Downloads         Music     Public      Templates       Videos
Documents  examples.desktop  Pictures  rootOnNVMe  test_write.img
jetson@jetson-desktop:~$ rm -rf /home/jetson/test_write.img
jetson@jetson-desktop:~$ ls
Desktop    Downloads         Music     Public      Templates
Documents  examples.desktop  Pictures  rootOnNVMe  Videos
jetson@jetson-desktop:~$
```

Note: Due to the storage characteristics of SSDs, increased storage space usage will affect read and write speeds. Therefore, try to test when there is plenty of free disk space.

Read and write speeds will vary from test to test. As long as the test data is not significantly different from the product specifications, it is considered normal. You can test multiple times to check the stability of the read and write speeds.